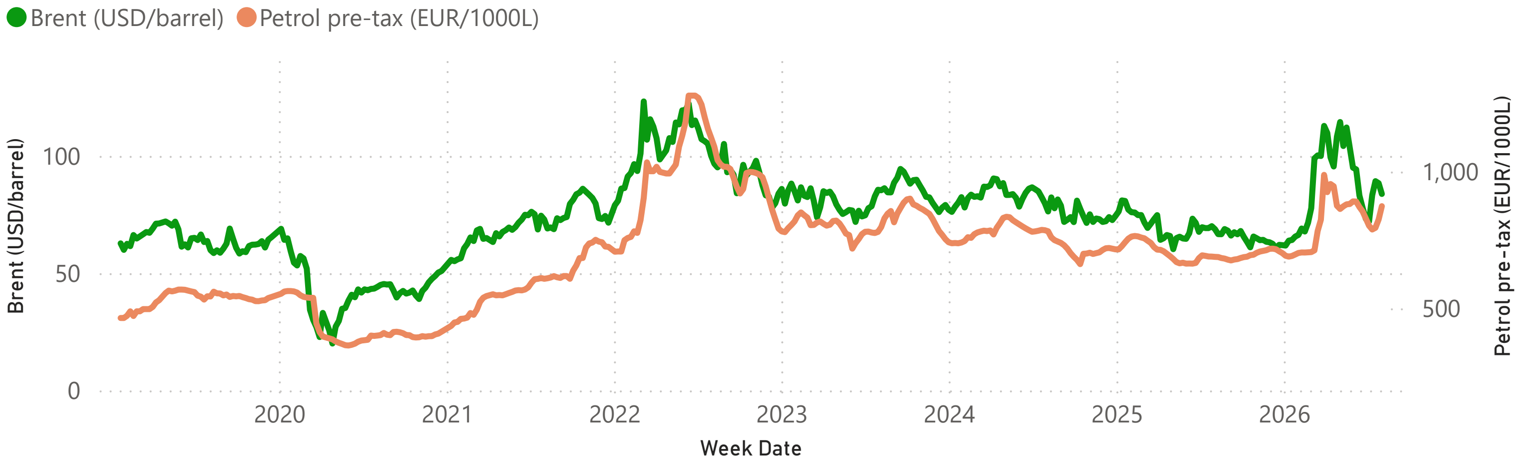
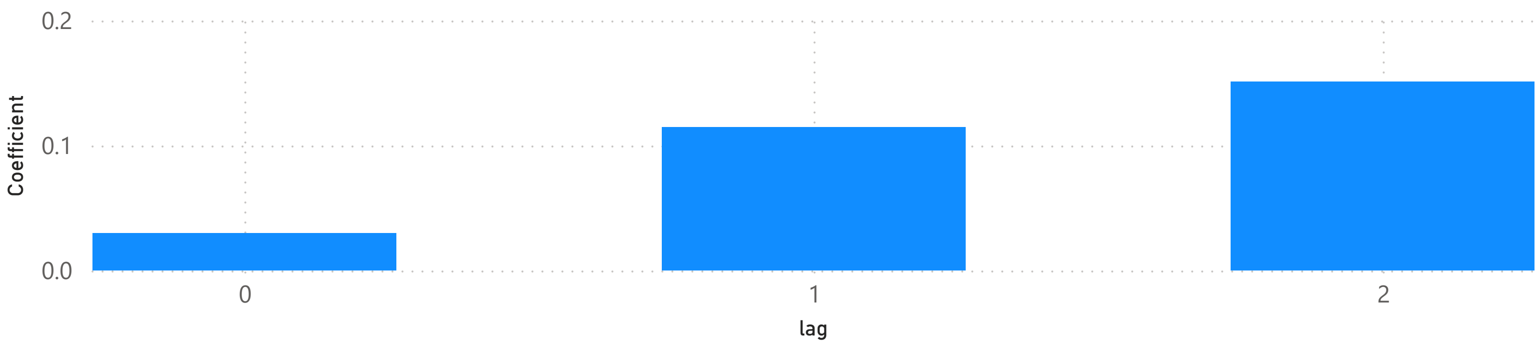
## Brent Crude Price (USD/barrel), 2019–2026




The graph shows a decline due to COVID-19, followed by a spike in 2022 tied to sanctions on Russian oil after the invasion of Ukraine, and another sharp spike in 2025–2026 driven by the Israel-Iran-US conflict, which disrupted Strait of Hormuz shipping. Pump prices broadly track Brent's trend, though the relationship isn't one-to-one — Brent explains roughly a quarter of week-to-week pump price movement, as shown in the pass-through analysis below.

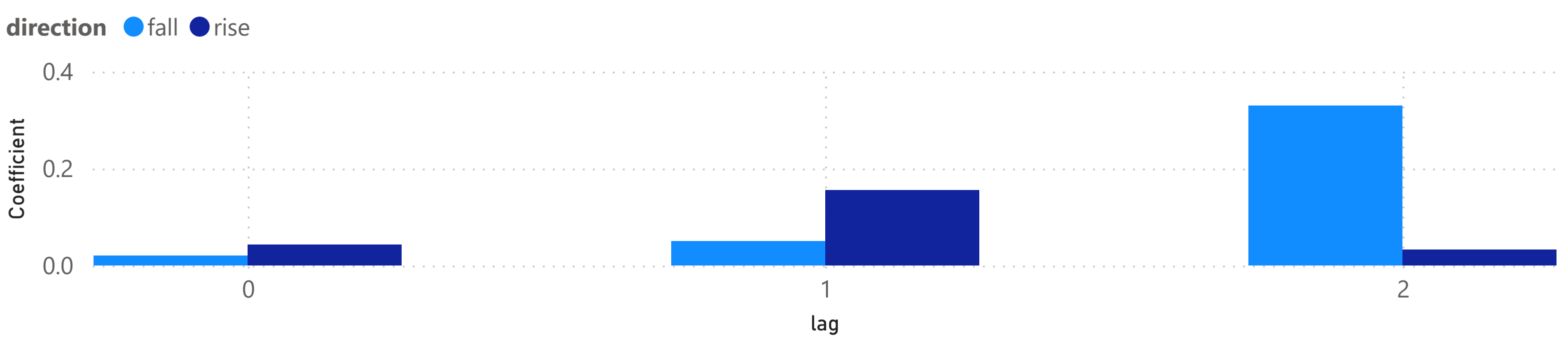
## Pass-through Effect by Lag (weeks)

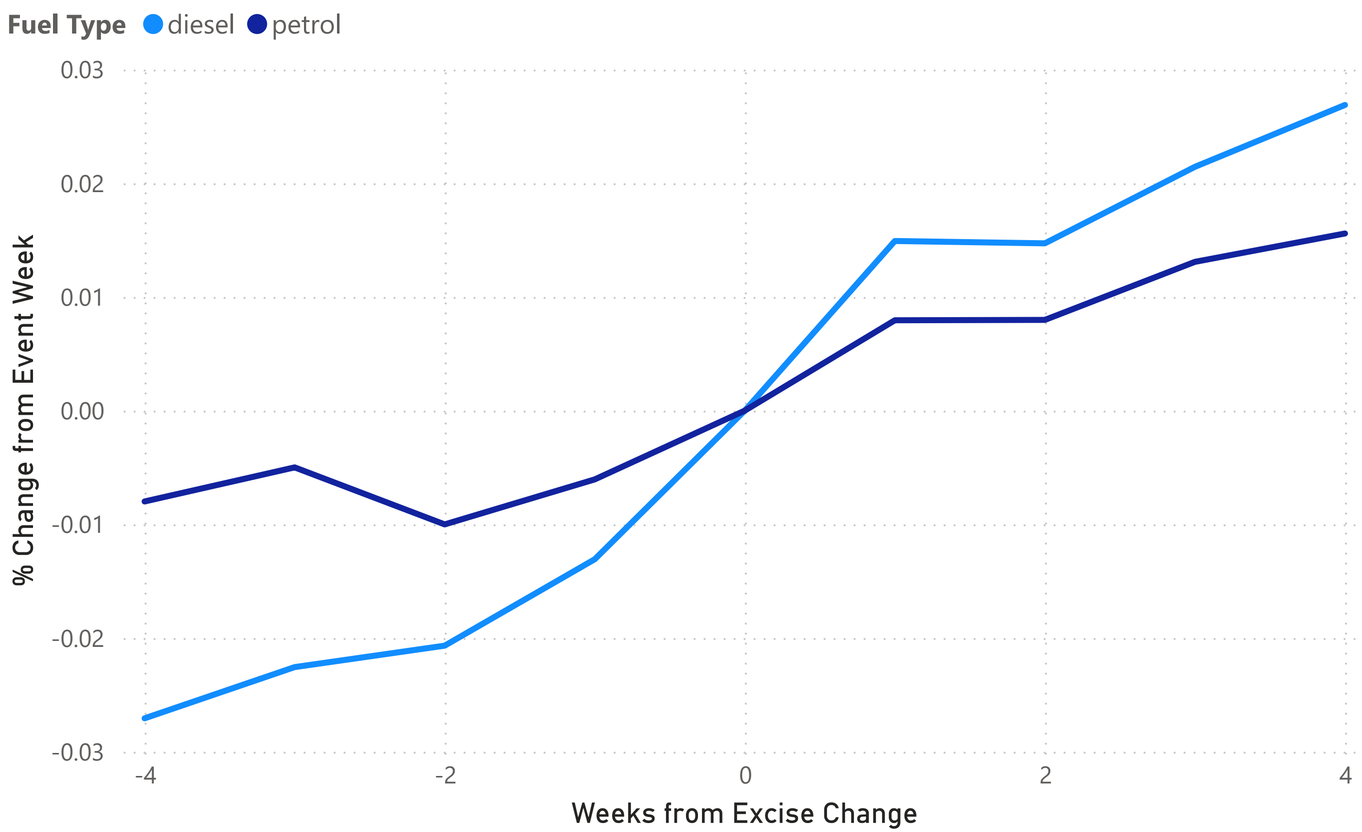



## Pass-through Effect by Lag

The price at the pump doesn't react to oil immediately—the reaction is spread out over time, and the full effect is only visible after 1-2 weeks. This is the substantive answer to research question #1 (pass-through rate)—not just "does Brent affect pump prices?" but "how quickly and how strongly?"—a concrete, measurable picture with a clear business interpretation (for example, for budget planning/hedging, you have about a week or two of head start before the oil price change is fully reflected in retail prices).

## Price Response: Rise vs Fall by Lag






## Price Response Around Excise Duty Changes

When the government raises fuel excise, how fast and how far does it show up at the pump? Looking at 16 petrol and 16 diesel excise changes, prices start drifting upward roughly a month before the change takes effect and keep rising for about a month after — not a sudden jump, a gradual shift.

Diesel moves more than petrol: a ~5.4pp swing across the window vs ~2.3pp for petrol, roughly twice as sensitive. Note this reflects the general price pattern around these dates, not a pure isolated tax effect — excise changes can coincide with broader oil price moves.